

Month 1

“Privacy-preserving Collaborative IDS for Internet of Vehicles using Federated Learning and Fully Homomorphic Encryption”

Scope: theoretical research, literature study, scheme survey, base-paper gap analysis, and current scheme selection.

1. Finding 1: Fully HE Is Impractical in Our Scenario

- **Latency issue:** IoVN intrusion detection requires real-time response times (10–100 ms) to mitigate safety-critical attacks like CAN bus injection, but FHE's intensive polynomial math and bootstrapping delays can stretch processing times to seconds or minutes, rendering immediate threat detection impossible.
- **Constraints On-Board Units (OBUs):** Vehicular embedded systems and OBUs are built for deterministic, low-power control operations rather than heavy cryptographic computation; running resource-heavy FHE algorithms drains vehicle battery power, causes thermal issues, and starves critical safety-essential applications of CPU/GPU cycles.
- **Ciphertext Size:** FHE expands standard 32-bit floating-point model weights into massive polynomial coefficients, resulting in gigabytes of data transmission that completely saturates highly dynamic, limited-bandwidth V2X (Vehicle-to-Everything) wireless networks.
- **Mobility Problem:** The rapid movement of vehicles causes frequent network handovers between different Roadside Units (RSUs); the extreme time required to perform an FHE-encrypted global model aggregation means vehicles will likely drive out of communication range before the process completes.
- **Incompatibility with Non-Linear IDS Activation Functions:** Effective deep learning-based IDS models rely heavily on non-linear functions (like ReLU or Sigmoid) to catch complex attack signatures, but because FHE can only natively perform addition and multiplication, approximating these functions requires deep polynomial circuits that trigger unsustainable computational overhead.

The practical implication is clear: if we try to encrypt everything, including unnecessarily deep computations, non-linear operations, or scheme-switching-heavy logic, the system will become too slow and too expensive to justify for a vehicular IDS.

2. Finding 2: HE Schemes and What Each Scheme Contributes

The month 1 literature review shows that there are several serious **homomorphic encryption families**. The final choice must be made based on the data type, the operation depth, the expected model size, and the vehicular deployment constraints.

(Note: Detailed Study still needs to be done over all these schemes individually to choose a scheme that is optimized for our scenario. Nevertheless, the following table presents a generic summary of the HE Schemes.)

Scheme family	Data type support	Strengths	Main limitation
BFV	Exact integers	Exact arithmetic; good for integer-valued statistics and database-style operations.	Requires quantization if weights are floating point; that adds extra error and engineering complexity.
BGV	Exact integers	Strong exact integer support; mature lattice-based family.	Same core issue as BFV: it is not naturally aligned with float-valued model weights.
CKKS	Approximate real/complex values	Best match for floating-point neural-network weights; supports SIMD packing and efficient weighted sums.	Approximation error must be managed carefully; parameter tuning is non-trivial.
TFHE / FHEW / CGGI	Bits / Boolean circuits	Excellent for boolean logic, comparisons, and fast bootstrapping in small circuits.	Bit-level evaluation is too expensive for large vector arithmetic and model aggregation.
Hybrid / scheme-switching designs	Mixed	Can combine the strengths of multiple schemes for different sub-tasks.	Scheme switching itself adds overhead and complexity; not ideal unless the workflow truly needs mixed operations.
Threshold / multi-key variants	Depends on base scheme	Useful when multiple parties must jointly decrypt or when key sharing is required.	Adds key-management complexity and is not needed for the basic secure-aggregation setup.

The scheme should be selected by matching the mathematics of the method to the mathematics of the model. Our model weights are real-valued, the main FL operation is weighted summation, and the target is privacy-preserving aggregation rather than arbitrary encrypted reasoning. That combination naturally favors **CKKS** (At least for now)

3. Finding 3: Biggest Gaps in the Base Papers

The base papers are useful, but month 1 review shows that neither paper fully solves the combined privacy, robustness, and deployability problem we care about.

Paper / category	What it achieves	What it still misses	Why that matters for us
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DCANFL	Decentralized FL with gossip-based collaboration and a continual-learning style update strategy.	Model updates are still exchanged without cryptographic protection; no strong defense against malicious peers or gradient leakage.	Privacy is incomplete, and peer-to-peer exchange makes secure key management harder.
FedloV	Robust centralized FL with TOPSIS + Multi-Krum filtering, encrypted transport, and adaptive client tuning.	The server still sees plaintext model updates during aggregation; the design is not protected against honest-but-curious inference on model updates.	Transport encryption alone is not enough; aggregation-time privacy is still open.
Common gap	Both papers improve collaborative IDS performance in different ways.	Neither paper gives cryptographic privacy for the model updates themselves, and neither fully addresses the cost of doing this securely in vehicular settings.	This is the exact opening where FHE fits.

4. Finding 4: Scheme and Parameter Selection So Far

Parameters fixed: 45,998 (As per the model architecture)

Scheme chosen: Based on the first month review, the current selected working direction is a particular form of CKKS which is known as “**levelled RNS-CKKS**” for encrypted model-update aggregation.

Why this choice is preferred right now:

- It matches the data type of the model (floating-point weights).
- It matches the main FL operation (summing / averaging updates).
- It avoids the extra quantization step required by BFV and BGV.
- It avoids the bit-level overhead of TFHE / FHEW for large vectors.
- It keeps the design aligned with the current month-1 goal: secure aggregation, not full encrypted inference.

(Note: This scheme is the current working selection, not the final decision.)